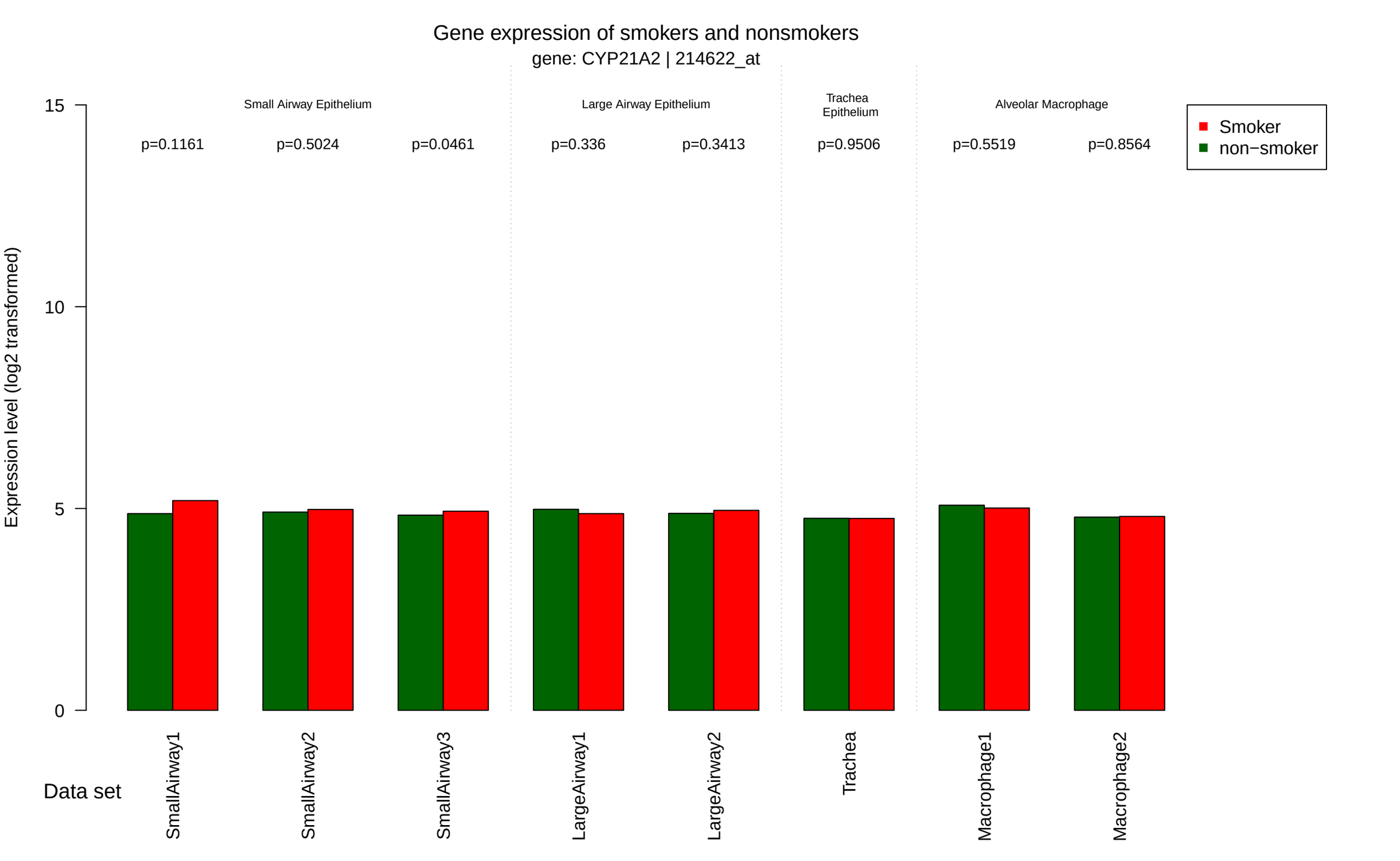
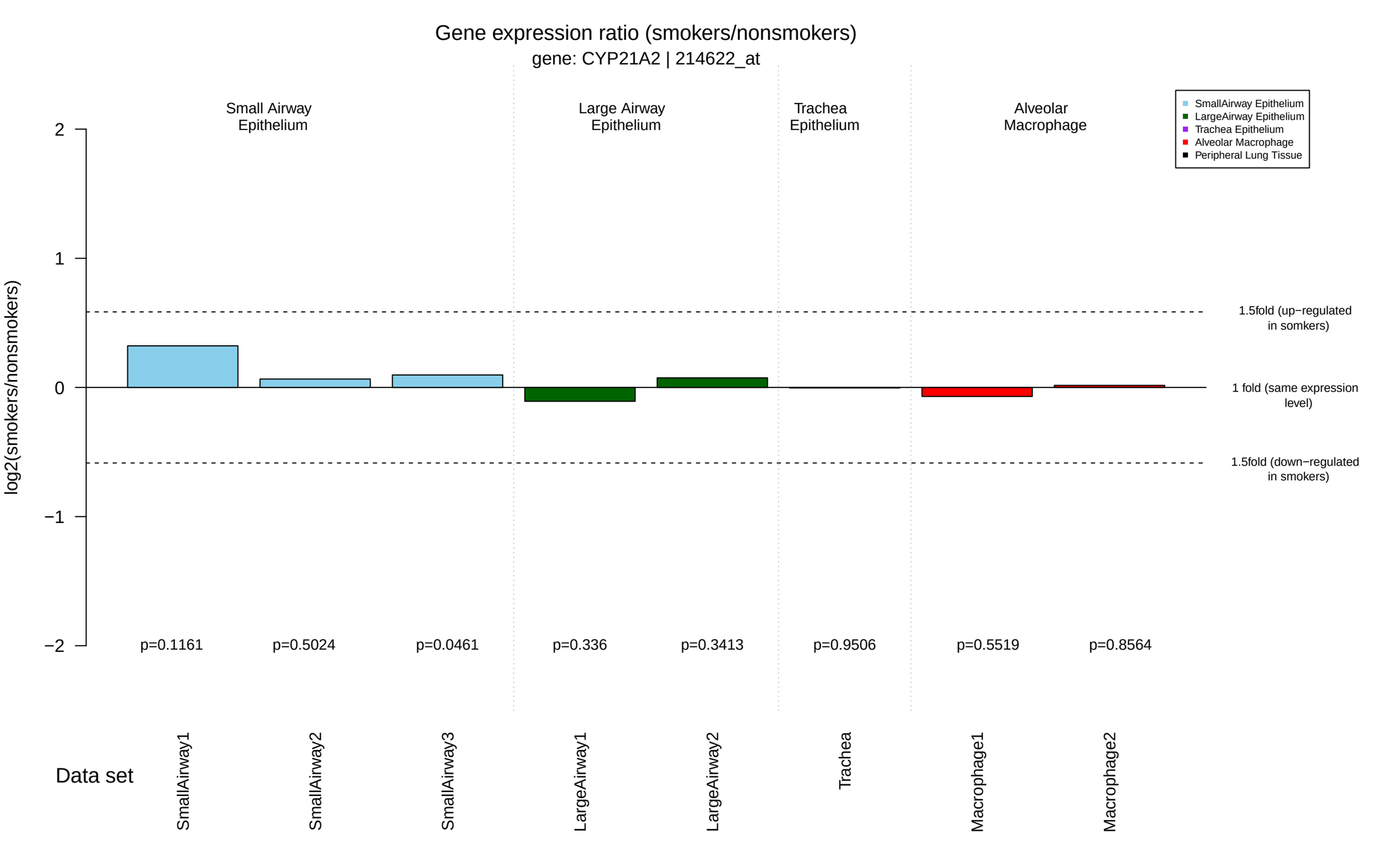
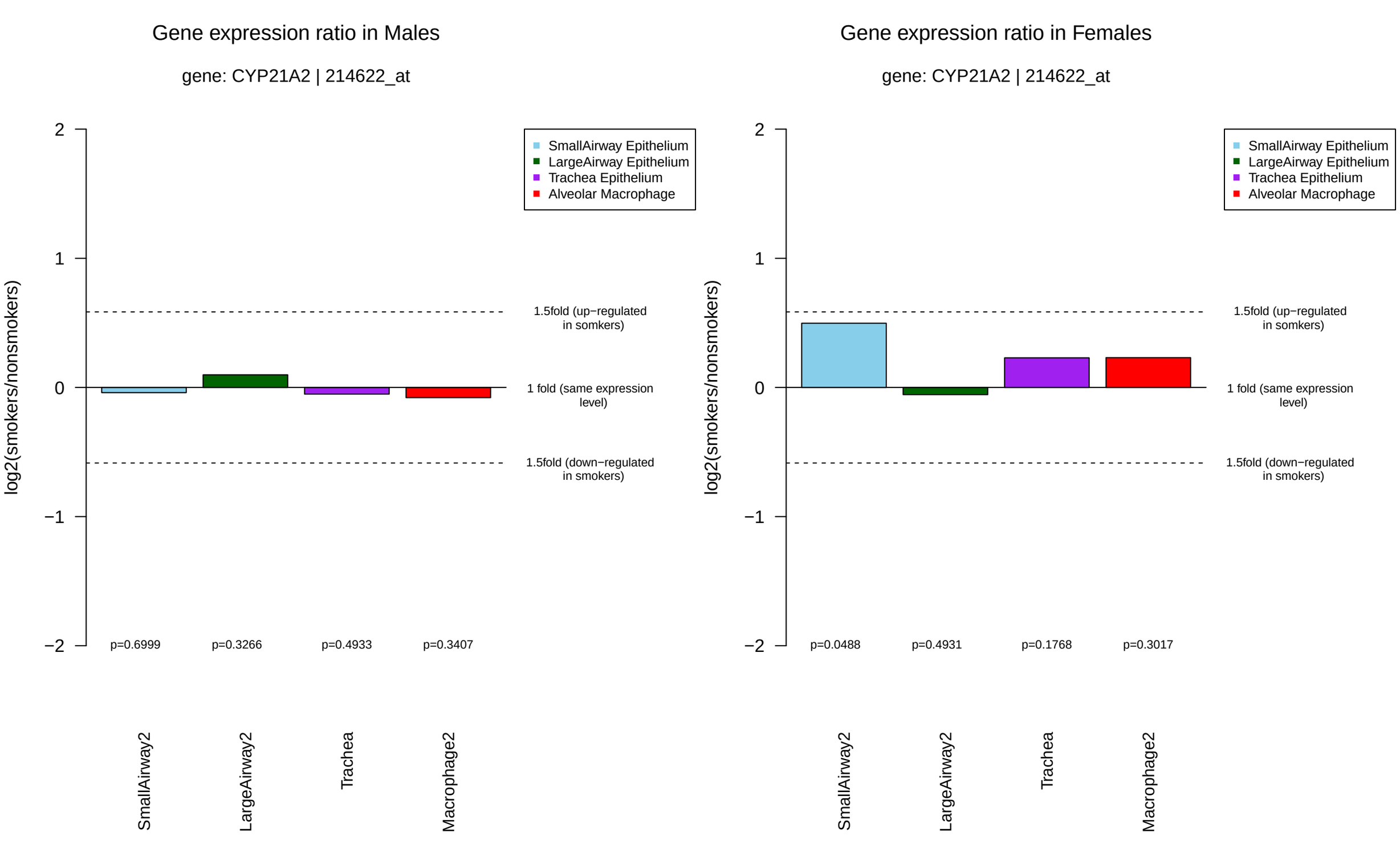
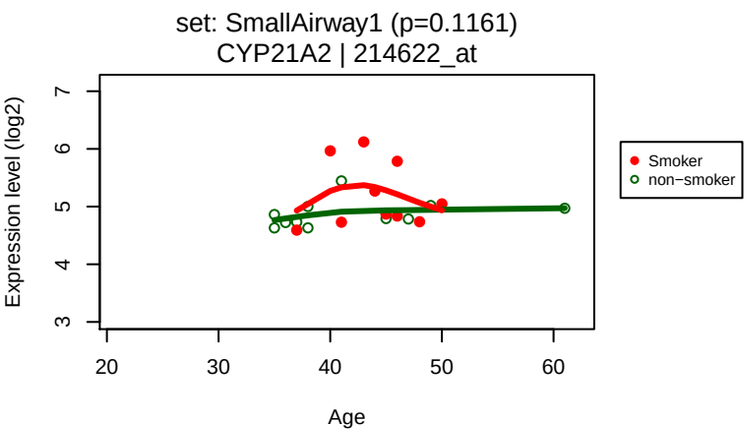
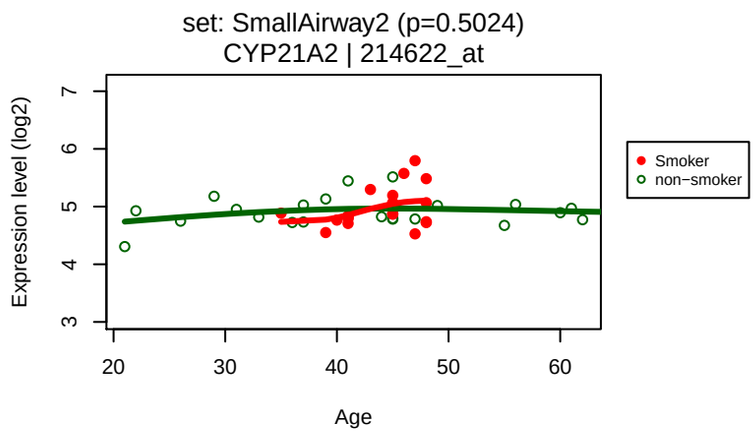
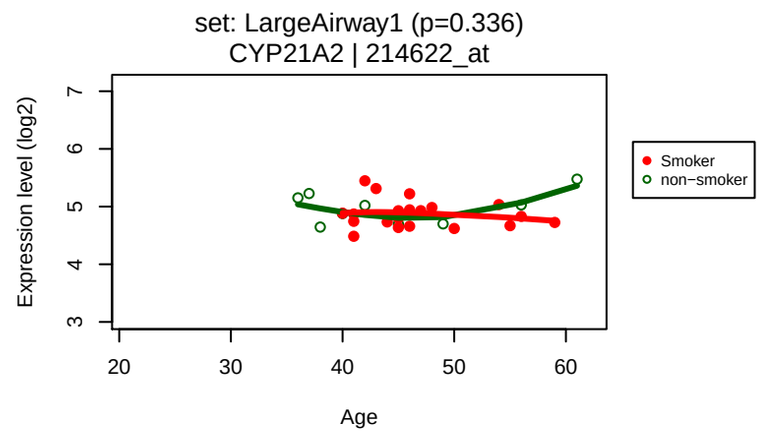

Gene expression of smokers and nonsmokers
gene: CYP21A2 | 214622_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1161
p=0.5024
p=0.0461
p=0.336
p=0.3413
p=0.9506
p=0.5519
p=0.8564
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

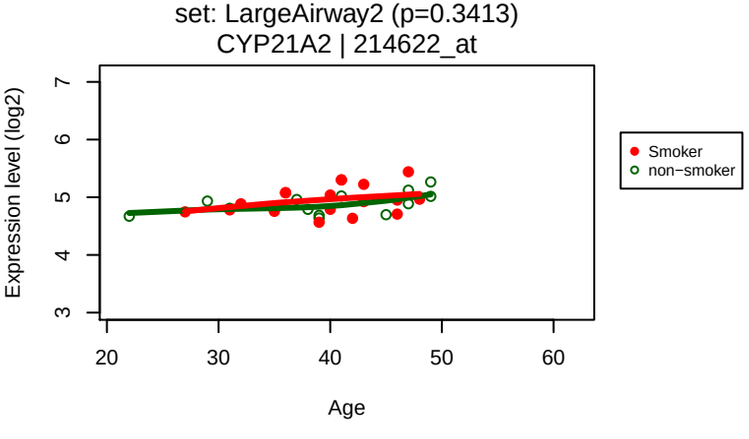

Gene expression ratio (smokers/nonsmokers)
gene: CYP21A2 | 214622_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1161
p=0.5024
p=0.0461
p=0.336
p=0.3413
p=0.9506
p=0.5519
p=0.8564
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

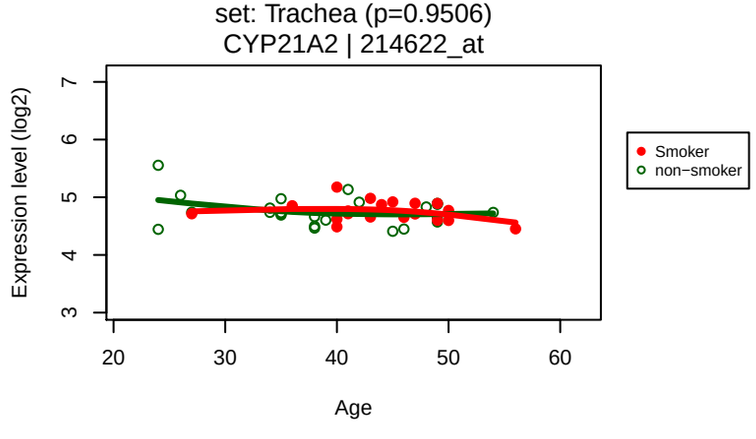

Gene expression ratio in Males
gene: CYP21A2 | 214622_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6999
p=0.3266
p=0.4933
p=0.3407
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP21A2 | 214622_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0488
p=0.4931
p=0.1768
p=0.3017
SmallAirway2
LargeAirway2
Trachea
Macrophage2

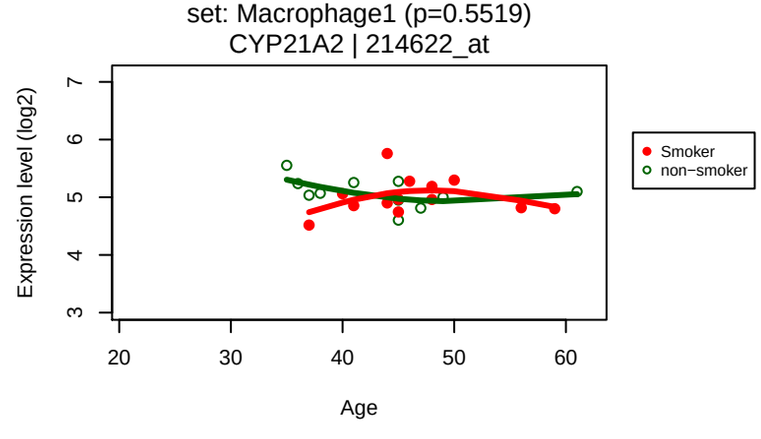

set: SmallAirway1 (p=0.1161)
CYP21A2 | 214622_at
Expression level (log2)
Age
Smoker
non−smoker

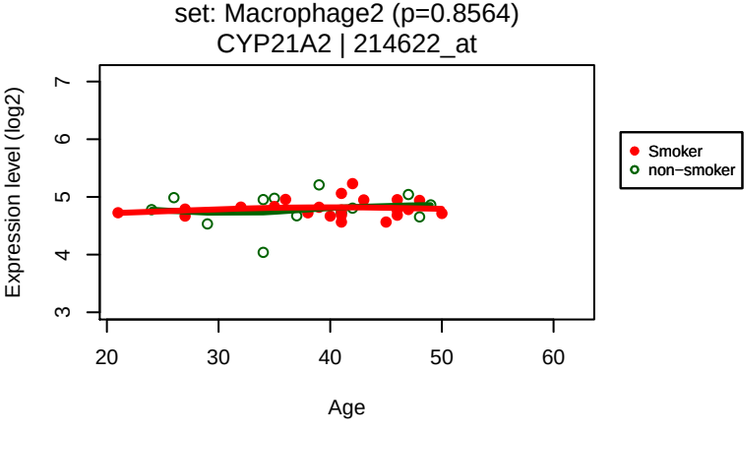

set: SmallAirway2 (p=0.5024)
CYP21A2 | 214622_at
Expression level (log2)
Age
Smoker
non−smoker

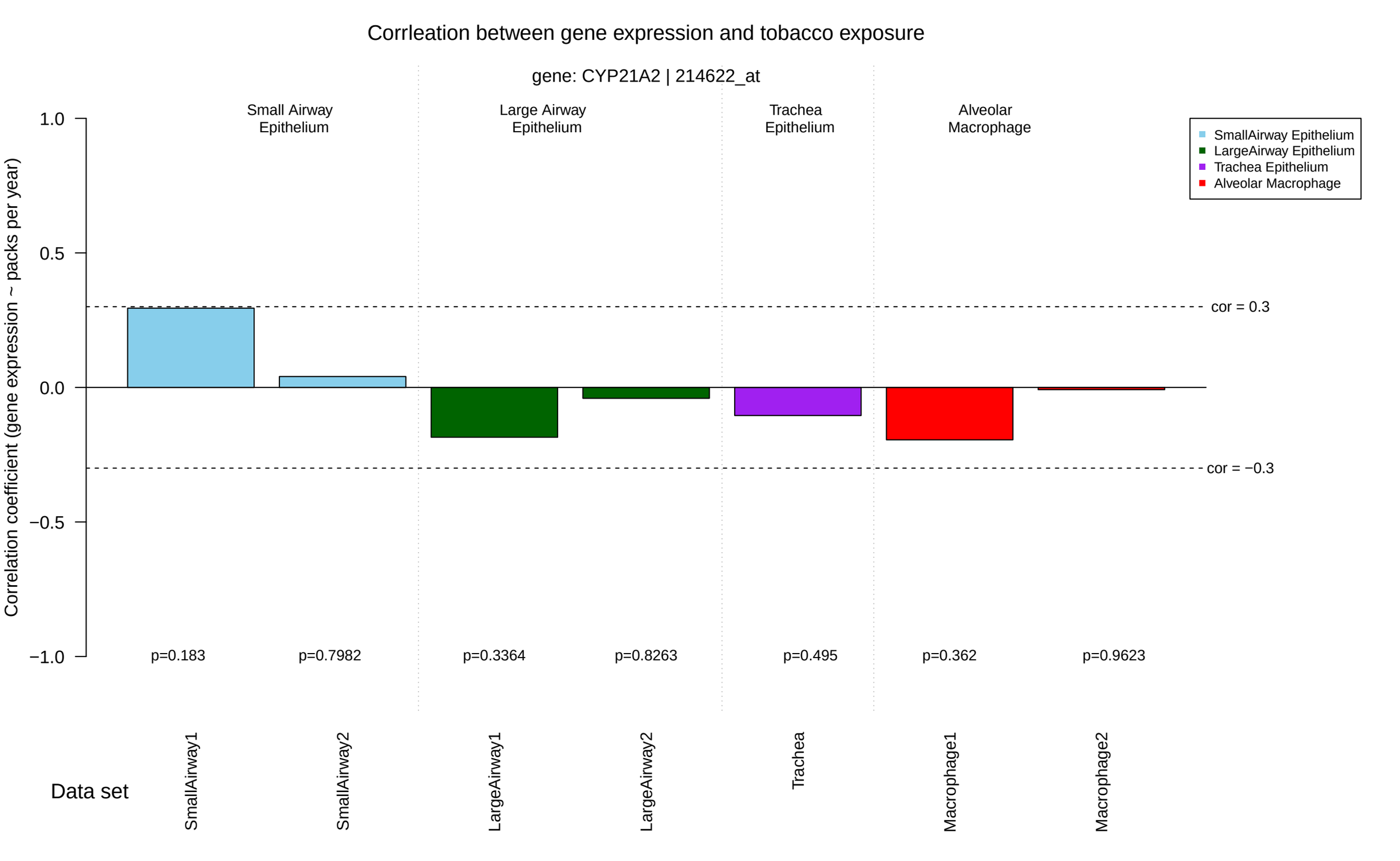

set: LargeAirway1 (p=0.336)
CYP21A2 | 214622_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3413)
CYP21A2 | 214622_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9506)
CYP21A2 | 214622_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5519)
CYP21A2 | 214622_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8564)
CYP21A2 | 214622_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP21A2 | 214622_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.183
p=0.7982
p=0.3364
p=0.8263
p=0.495
p=0.362
p=0.9623
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2